Detection of Word Shift Steganography in PDF Document

Li Lingjun*, Huang Liusheng, Yang Wei, Zhao Xinxin, Yu Zhenshan, Chen Zhili
1. Depart of C.S. & Tech., NHPCC, USTC, Hefei, 230027, P. R. China
2. Suzhou Institute for Advanced Study, USTC
* ljli@mail.ustc.edu.cn

ABSTRACT

Word shift is a fundamental format based text steganography. It embeds secret information in text by shifting words slightly. Compared with study on steganography, research on its steganalysis is still in its infancy. In this paper, we present a blind steganalysis method to detect word shift in PDF document. Our method is to find features sensitive to word shift and use classifier to learn and remember feature differences between natural document and stego one. In order to design sensitive features, we propose two concepts "neighbor difference" and "environment equal", which reveal the spaces' statistical property. Then, we divided the PDF document into two types to make our method work efficiently. At last, we design a series of experiments to demonstrate performance of our method. The detection accuracy of our method can be up to 93.3%. Our initial results shows that our proposed concepts are very useful in text steganalysis and offer help for other related works.

Categories and Subject Descriptors

C.2.0 [**Computer-Communication Networks**]: General—*Security and protection*

General Terms

Security

Keywords

text steganography, steganalysis, information hiding, word shift, PDF

1. INTRODUCTION

Today, text-based information exchange or distribution on Internet, like accademic paper, email, e-book and so on, plays an increasingly important role in people's daily life. As an excellent format document, PDF document is the prevalent carrier of these information. The fact that huge number of PDF documents are transmitted on Internet increases concern on using text in PDF document as a covert channel of communication. One of such important covert method is known as format based steganography. Its main idea is to embed secret information by modifying format structure, like word position, character color, serif and etc., so that the resulting stego-document would seems like the normal covert document and the secret information can not be obtained by anyone other than the intended recipient.

Word shift steganography is one of the most important format based methods. Because it does not alter text's original meaning and works quickly and simply, word shift is widely used. It shifts the word position, or in other words change the inter word space length, to embed secret message. For instance, hider widens space to denote bit "1" and shrink it to denote bit "0". A simple example is shown in Fig. 1

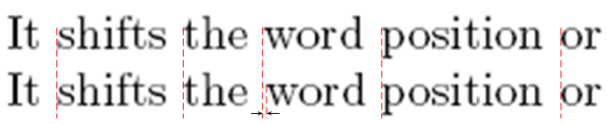


Figure 1: Simple example of word shift steganography. The top line is normal text. The bottom line widens the space before "word" to hide bit "1".

When extracting secret information, the method above needs to compare spaces in received text with ones in original "pure" text. Brassil et al. proposed an idea of "controlling space" in Ref. [1], which makes the stego text become self extracted. It assumes that spaces in a row are equal to each other. So, algorithm keeps the embedding position's neighbor spaces (before and after) unchanged during embedding phase. These two spaces are called "controlling spaces". In the extraction phase, algorithm compares embedding position with controlling spaces to determine weather the space at embedding position is widened or shrunk. However, this method notably alters a text line length.

Based on the Brassil's idea, Chen at el. proposed a new algorithm in Ref. [2]. This algorithm slightly changes the total length of a text line, which increases the security and practicability of word shift steganography. Similar to Brassil's, this algorithm also uses controlling spaces. Difference occurs in how to embed bits. Hider change the space length at embedding position as follows:

1. Initialize a counter: $C = 1$
2. Find next embedding position.
   - IF all positions are used out. Go to 5

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- IF a line is finished, $C = 1$

3. Find next bit to be embedded.
   - IF there is no bits. Go to 5.
   - IF the bit is "0", keep the embedding position constant, go to 2.
   - IF the bit is "1", go to 4

4. IF C is an odd number, shrink the space. ELSE, widen the space. $C = C + 1$. Go to 2.

5. End Algorithm

This algorithm makes the steganographic capacity up to half the number of total spaces and will not break the alignment of each line.

In Ref. [3], Zou et al. gave a word shift watermark algorithm robust to printing, copying and scanning. It firstly divides spaces in a text line into two sets, Set A and Set B. The number of spaces in each set are equal. The division method is that first half of spaces in a line are grouped into set A, and the second half into set B. Then, if bit "0" is coded, hider widen spaces in set A evenly and shrinks ones in set B to make total length of text line unchanged and $\Phi_A - \Phi_B = \xi$, where Φ_A, Φ_B are the length summation of spaces in A and B, and ξ is called embedding strength in the paper. It is similar to embed "1" with shrinking set A and widening set B to make $\Phi_A - \Phi_B = -\xi$. This algorithm is robust yet low in capacity.

In extracting process, in order to detect word shift or measure word space length, a vertical profile of words is usually compiled from a digitalized page image [4], [5]. Suppose $f(x, y) \in [0, 1]$ represent the grayscale at position (x, y) of a page image, the vertical profile of words

$$v(x) = \sum_{y=t}^{b} f(x, y), \quad x = 0, 1, \cdots, W$$

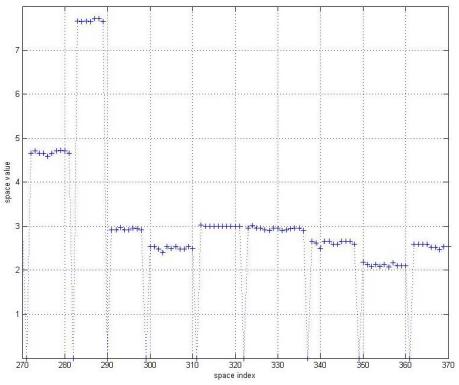
is the sum of grayscale along the vertical scan-lines x [5]. Here, W is the width of a page and t,b are top and bottom "boundaries" of a text line. The figure of vertical profile is consist by short columns and narrow valleys. These valleys are the inter word spaces. Due to existence of profile noise, some methods are taken to eliminate the effect of noise distortion. The correlation detection and centroid detection are two main methods. These are traditional measurement used in the areas of page image oriented word shift steganography. However, as we will focus on digital PDF document in this paper,the space measurement can be simplified. Page image can be converted to formatted text by using Adobe ORC (optical character recognition) software and word positions can be obtained from PDF file objects. In PDF imaging model, each symbol has a corresponding coordinate, which is stored in *text matrix* and *line matrix*, two PDF objects in PDF file. After obtaining every word's position coordinate value, inter word space length can be calculated out. Yet, this calculating work is complicate and tedious [6]. Fortunately, we find PDFlibTET library [7], a professional PDF program interface, to help us get the coordinate value.

Compared with the steganography, steganalysis of word shift method is little discussed. Ref.[8] gives an analysis on methods, which hide bits in binary text image, by using idea of soft pattern matching. It pairs the similar marks extracted from the binary image and then compares them to determine weather hidden information exists and to estimate the size of hidden information. Its estimating error is between -15% and 15%. Ref. [9] breaks open space method by investigating the ratio of Characters to spaces. Ref. [10] uses Support Vector Machine (SVM) to break some kinds of format based steganography. But, their selected features are not suitable for detecting word shift algorithm because they can not reflect the complicate characteristic of text word space.

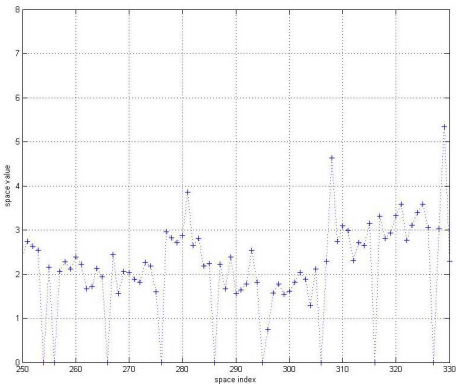
1.1 Outline of this paper

There are two types of steganalysis: algorithm-specific method and blind method. The former breaks a specific steganographic method [11] and the latter can deal with a class of steganographic algorithms. In this paper we design a blind detector for a class of word shift steganography in PDF document. Using our detector, we can find weather the words in a PDF document have been maliciously shifted or not.

The rest of paper is organized as follows: First of all, we establish a two-level detection model, described in Sec. 2. In Sec. 3, we give a detailed description for features specific to different classifiers. A series of experiments are shown and their results are discussed in Sec. 4. Sec. 5 gives the conclusion of this paper.



(a)



(b)

Figure 2: Word space of two types of PDF documents. (a): type I (b): type II

2. DETECTION MODEL

In a cover document, word spaces are determined by three components ([10]): stationary signal P_i, implicating the space's original length, noise signal δ_i, caused by conversion, and stego signal γ_i, representing the secret bits. So, the ith space is:

$$S_i = P_i + \delta_i + \gamma_i \quad (1)$$

In a natural text line, P_i equals each other and γ_i is 0.

We find that all PDF documents can be divided into two types according to the way word spaces change (Fig.2). This is due to two different producing approaches. Type I is produced by word processing or typography software, such as Microsoft Word and LaTeX. We call them soft documents. In this type, there is no noise pollution, $\delta_i = 0$. Type II is converted from real paper image. They are firstly scanned into the computer and then turned to digital PDF document by using ORC software. We call them hard document. The noise in this type leads δ_i having a random value. Fig. 2 shows some lines of natural word spaces in two types of documents. X-axis is index of word spaces and Y-axis is their values. Value 0 denotes line break. From the figure, it can be seen that space length of type I document nearly equal each other in a line. Their differences, if exists, are less than 0.1 pt. Oppositely, space length changes dramatically in type II document. Since we do not know this change is caused by noise or by artificial modifying, it is difficult to detect stego documents created from type II.

Our detection model firstly classifies unknown document into one of two types (using classifier 1) and then extracts different features specific to different document type in order to judge weather the document is a stego one (using classifier 2 and 3). Fig. 3 shows our detecting process. Feature

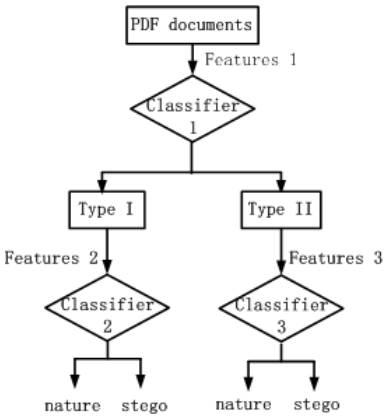


Figure 3: Our detecting process

1, feature 2 and feature 3 are extracted from documents. Then, we train the classifiers with sample documents. The detection efficiency depends on the features and the training samples. After training, the system can work for detection. In this paper, we select Support Vector Machine (SVM) as classifier because of its outstanding performance.

3. CALIBRATED FEATURES

In our detect scheme, we use two types of features — first order features and high order features. The former is designed for simple classifier 1 and 2 and the latter is employed in complicate classifier 3.

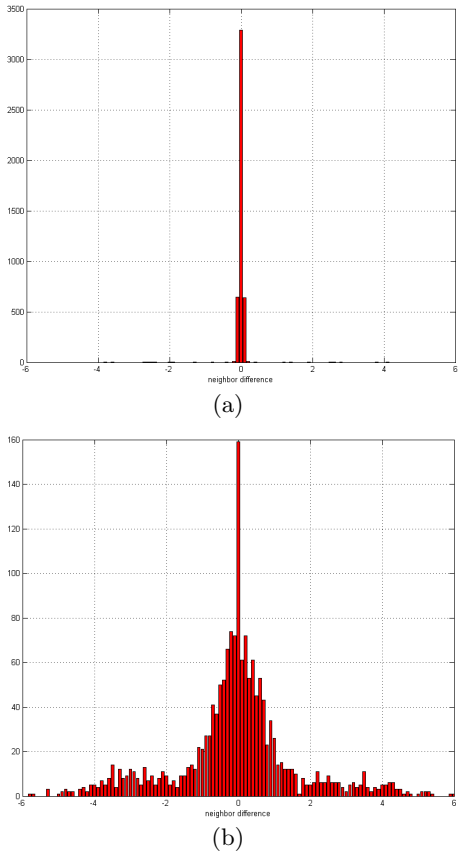


Figure 4: Histogram of Neighbor Difference, measured in unit pt ($\frac{1}{72}$ inch). (a): type I (b): type II.

3.1 Classifier 1

Classifier 1 is used to determine which type the unknown document belongs to. We make use of *neighbor difference* D_i, which is defined as difference of two successive spaces:

$$\begin{aligned} D_i &= S_i - S_{i-1} \quad (2) \\ &= P_i - P_{i-1} + \delta_i - \delta_{i-1} + \gamma_i - \gamma_{i-1} \end{aligned}$$

Suppose $\{D_i\}_{i=1}^N$ denotes the neighbor difference set of natural document. Then its histogram $H_k^{\mathbb{D}}$ means the number of elements D_i whose value equal k. $H^{\mathbb{D}}$ of two types of documents are different, which are shown in Fig. 4. It can be seen that type II's histogram arranges more widely than type I's. Histogram variance reflects this difference. So feature 1 is variance of $\{D_i\}_{i=1}^N$ (Equ. 4):

$$\bar{D} = \frac{1}{N}\sum_{i=1}^{N} D_i \quad (3)$$

$$S_D^2 = \frac{1}{N-1}\sum_{i=1}^{N}(D_i - \bar{D})^2 \quad (4)$$

3.2 Classifier 2

The job of Classifier 2 is to detect existence of secret message in type I documents. From Fig. 2(a), it can be seen that in a text line of type I, word spaces nearly equal each other. Fig. 4(a) shows that the change of spaces is limited

in range $[-0.01, 0.01]$. Therefore, any artificial word shift will result in noticeable alteration in histogram, represented by $H_k^{\mathbb{D}}$, where $k = \min_i D_i, \cdots, \max_i D_i$. Its frequency is defined as:

$$F_k^{\mathbb{D}} = \frac{H_k^{\mathbb{D}}}{\sum_i H_i^{\mathbb{D}}}, \quad k = \min_i D_i, \cdots, -0.01, 0, 0.01, \cdots, \max_i D_i \tag{5}$$

We choose $[F_{-0.1}^{\mathbb{D}}, F_{-0.09}^{\mathbb{D}}, \cdots, F_{0.09}^{\mathbb{D}}, F_{0.1}^{\mathbb{D}}]$ as final feature vector of classifier 2, because most of natural neighbor difference gather in this range.

3.3 Classifier 3

Compared with above two detections, our biggest problem is how to detect secret message in type II. The detection difficulty is due to distortion introduced by noise component γ (Fig. 2(b)) in conversion. Our main idea is to find high order features sensitive to artificial shift yet ignorant to noise distortion.

3.3.1 neighbor difference

As shown in Equ. 2, steganographic algorithm alters the stego component $\delta_i - \delta_{i-1}$ from 0 to $\pm\mu$. Here, μ is the value added to original space length. Suppose $\{D_i'\}_{i=1}^{N'}$ denote the stego neighbor difference set, then relationship between its histogram $H_k^{\mathbb{D}'}$ and natural one $H_k^{\mathbb{D}}$ is:

$$H_k^{\mathbb{D}'} = (1-R) \times H_k^{\mathbb{D}} + \frac{R}{2} \times H_{k-\mu}^{\mathbb{D}} + \frac{R}{2} \times H_{k+\mu}^{\mathbb{D}} \quad k = \cdots, -0.01, 0, 0.01, \cdots \tag{6}$$

where R is embedding rate, that is the ratio of embedding position to total spaces, and k is the index of histogram. Equ. 6 means that embedding process transforms space length into others whose difference is $\pm\mu$, just like PoVs in image steganograhy[12]. In histogram (Fig. 4(b)), word shift will shorten the peak at value 0 and create two little peaks at $\pm\mu$ separately. Higher bars will descend while lower ones will increase. Four features have been selected.

mean and variance: The mean of neighbor difference $\bar{D}$ (c.f. Equ. 4) is near 0 because histogram is symmetrical about value 0. However, word shift will break this symmetry if the algorithm does not choose embedding positions exactly. On the other hand, Variance S_D^2 (c.f. Equ. 4) reflects the dispersion of point D_i around mean $\bar{D}$. Word shift drives point D_i distributes away from mean, which increases S_D^2.

frequency of value 0 ($F_0^{\mathbb{D}}$): In the set $\{D_i\}_{i=0}^N$, value 0 has the highest frequency of occurrence. This frequency will notably change as soon as word spaces have been modified. So it makes sense to choose it as a feature

$$F_0^{\mathbb{D}} = \frac{H_0^{\mathbb{D}}}{\sum_{\{D_i\}_{i=1}^N} D_i} \tag{7}$$

K-L divergence: The histogram of neighbor differences (Fig. 4(b)) approximately follows Laplase Distribution, whose probability density distribution is written as:

$$p(x|\mu, b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right) \tag{8}$$

cumulative distribution function is :

$$F(x) = \frac{1}{2} \exp(-\frac{|\mu - x|}{b}) \tag{9}$$

mean is μ and variance is $2b^2$. Laplace distribution can be used to estimate frequency set of natural neighbor differences $\{F_k^{\mathbb{D}}\}_{k=\cdots,-0.01,0,0.01,\cdots}$ (c.f. Equ. 5). Using maximum likelihood estimation, expected mean $\hat{\mu}$ is the median of set $\{D_i\}_{i=0}^N$. Suppose the set is independent and identically distributed, expected $\hat{b}$ is

$$\hat{b} = \frac{1}{N} \sum_{i=1}^{N} |D_i - \hat{\mu}|$$

Using Equ. 9, expected frequency of natual neighbor differences is obtained:

$$\hat{F}_k^{\mathbb{D}} = (F(k+0.005) - F(k-0.005)) \quad k = \cdots, -0.01, 0, 0.01, \cdots \tag{10}$$

The extent of similarity between the two frequency set, $\{F_k^{\mathbb{D}}\}$ and $\{\hat{F}_k^{\mathbb{D}}\}$, determines whether the document is like a natural one. This similarity is measured by K-L divergence of actual set $\{F_k^{\mathbb{D}}\}$ from expected natural set $\{\hat{F}_k^{\mathbb{D}}\}$:

$$D_{KL} = \sum_k \hat{F}_k^{\mathbb{D}} \cdot \log \frac{\hat{F}_k^{\mathbb{D}}}{F_k^{\mathbb{D}}} \tag{11}$$

The smaller the D_{KL} is, the more similar the document is to a natural one. Compared with above two features, K-L divergence supplies more detailed information about distribution of neighbor differences.

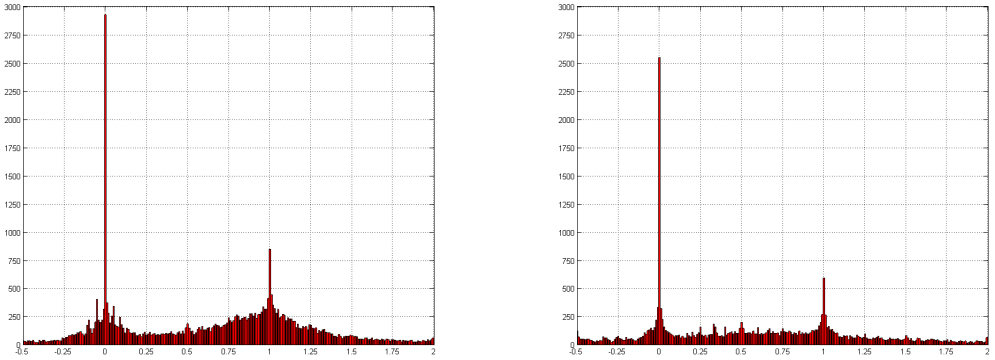
3.3.2 environment equal

As we mentioned above, type II document is converted from paper page, so its shift value μ is usually "distorted" by noise during the progress of printing and digitalizing. High order features are required to resist this distortion. Here, we introduce the idea of "*environment equal*".

Word space locates between two letters — the last one of prior word and the first one of next word. In type II document, space length is related to these two letters. For example, if the spaces are all the same in soft documents, after they are printed and converted to PDF documents, we find that most spaces between letters (m,m), will be narrower than those between (l,l). Let us focus on the triples $T_i^l = (p_i, S_i^l, n_i)$, where $p_i(n_i)$ is the letter prior(next) to space S_i^l and l is the line index. The prior letter and the next letter consist a space's "environment". We say two triples T_i^l and $T_{i'}^{l'}$ are *environment equal* when $(p_i, n_i) = (p_{i'}, n_{i'})$. Consider new data set $\mathbb{Q}$:

$$\mathbb{Q} = \left\{ Q_j = \frac{S_{i_1}^{l_1} - S_{i_2}^{l_2}}{S_{i_3}^{l_1} - S_{i_4}^{l_2}} \;\middle|\; (p_{i_1}, n_{i_1}) = (p_{i_2}, n_{i_2}), \; (p_{i_3}, n_{i_3}) = (p_{i_4}, n_{i_4}) \right\} \tag{12}$$

During conversion process, environment equal spaces will receive similar degree distortion. In another words, if these spaces are same in soft document, they will have little difference after conversion. In soft document, spaces are considered to equal each other in the same line, but it is possibly not the case when they are in different lines. To get rid of



(a) (b)

Figure 5: (a): Histogram of natural set $\mathbb{Q}$ (b): Histogram of stego set

this difference, Equ. 12 selects lines which have two environment equal triples with other lines and gets the ratio of the space difference of the two environment equal triples.

A typical histogram of set $\mathbb{Q}$ is shown in Fig. 5(a). There are two peaks. The higher one locates at value 0 because space length in most environment-equal triples are same. Peak at value 1 is formed by same level of noise disturbance between two environment-equal triples. Most data in $\mathbb{Q}$ distribute around these two peaks. Similar to influencing histogram of set $\mathbb{D}$, a steganographic process also alters this histogram. A stego instance is shown in Fig. 5(b). We select four high order features relevant to $\mathbb{Q}$:

Frequency of value 0 ($F_0^{\mathbb{Q}}$): similar to the histogram of $\mathbb{D}$, space modifying will dramatically change the height of the bar of value 0.

$$F_0^{\mathbb{Q}} = \frac{H_0^{\mathbb{Q}}}{\sum\limits_{\{Q_i\}_{i=1}^{N}} Q_i} \tag{13}$$

***var*$_0^{\mathbb{Q}}$ and *var*$_1^{\mathbb{Q}}$**: the extent of data gathering around two peaks (0 and 1) is an effective indication for distinguishing natural document from stego one. This extent can be measured by two features below:

$$var_0^{\mathbb{Q}} = \frac{1}{\sum\limits_k H_k^{\mathbb{Q}}} \sum_k H_k^{\mathbb{Q}} \cdot k^2$$
$$k = -0.25, \cdots, 0, \cdots, 0.25 \tag{14}$$
$$var_1^{\mathbb{Q}} = \frac{1}{\sum\limits_k H_k^{\mathbb{Q}}} \sum_k H_k^{\mathbb{Q}} \cdot (k-1)^2$$
$$k = 0.25, \cdots, 0, \cdots, 1.5 \tag{15}$$

Like variance in statistics, $var_0^{\mathbb{Q}}$ and $var_1^{\mathbb{Q}}$ reflect dispersion of data near 0 and 1. Data in [0,1](Fig. 5(a)) is divided into two groups. From the experience, we find that point 0.25 would be a good boundary of these two groups.

Frequency of similar spaces:As we have analyzed, the environment equal triples in a same hard document line would be little different. Let set $\mathbb{P}$ contain environment equal spaces of each line:

$$\mathbb{P} = \left\{ (S_i^l, S_j^{l'}) \mid (p_i, n_i) = (p_j, n_j) \text{ and } l = l' \right\}$$

then select "similar" space out of set $\mathbb{P}$

$$\mathbb{P}_{\text{similar}} = \{(S_i, S_j) \mid (S_i, S_j) \in \mathbb{P} \text{ and } |S_i - S_j| < \delta\}$$

δ is a tolerance value. Frequency of the similar spaces can be used as our final feature:

$$F_{\text{similar}} = \frac{\|\mathbb{P}_{\text{similar}}\|}{\|\mathbb{P}\|} \tag{16}$$

where $\|\mathbb{P}\|$ is the number of elements contained in set $\mathbb{P}$.

At the end of this section, all the selected features are listed in the Tab. 1

Table 1: All selected features

Classifier 1	S_D^2
Classifier 2	$F_{-0.1}^{\mathbb{D}}, F_{-0.09}^{\mathbb{D}}, \cdots, F_{0.09}^{\mathbb{D}}, F_{0.1}^{\mathbb{D}}$
Classifier 3	$\bar{D}$, S_D^2, $F_0^{\mathbb{D}}$, D_{KL} $F_0^{\mathbb{Q}}$, $var_0^{\mathbb{Q}}$, $var_1^{\mathbb{Q}}$, F_{similar}

4. EXPERIMENT RESULTS

We have designed several experiments to test three classifiers separately and put most of our attention on the efficiency of classifier 3. We use LIBSVM as our classifier [13].

Before experiments, we need to describe the way to create experimental PDF documents. Natural type I document is converted from natural Word document, which is randomly downloaded from Internet. We download nearly 7,000 Word pages. Natural type II document is downloaded from IEEE online library, where many electric documents are scanned from real papers. We use a program to produce stego Word document by using Microsoft Word program interface. Two main parameters, shifting distance μ and embedding rate R (see Sec.3.3.1), are different in different tests. According to the given R, we randomly select the embedding position and widen the space by μ pt. Then we convert result stego Word document to PDF document. This creates stego type I document. The way we produce stego type II document will be described later.

To test classifier 1, we download 3817 pages (in 400 documents) type II PDF documents from IEEE online library and create another 4000 pages (in 400 documents) type I

document. Compute their variances of neighbor difference histogram and input them into classifier 1. 200 type I documents and 200 type II documents are used as training samples and remaining documents are testing samples. The experiment result is shown in followed two-by-two confusion matrix (Tab. 2). This confusion matrix means that in 200

Table 2: confusion matrix of classifier 1

		True Type	
		I	II
Classified as	I	197	8
	II	3	192

type I testing samples, 197 are classified correctly and 3 are not. 192 type II testing samples are given right label and remaining 8 are not. The efficiency of classifier 1 is satisfied for our next detecting steps.

We create 100 natural type I documents and 100 stego type I documents to train classifier 2. To test its performance, we create 5 groups with 100 stego documents in each group. The embedding rate R of these groups are: 5%–25% in 5% increment and μ of each space is a random value in the range [-1,1]. Also, we produce another 100 testing natural type I documents. The experiment result is shown in Tab. 3 The high detection accuracy of classifier 2 is owing

Table 3: detection result of classifier 2

	nature	5%	10%	15%	20%	25%
success	93	94	97	96	98	96
fail	7	6	3	4	2	4
accuracy	93%	94%	97 %	96%	98%	96%

to simple histogram of type I neighbor difference.

Before testing classifier 3, we introduce our substitute method of creating real stego type II document. To get a real one requires hider to shift words in a natural PDF document, printing document on papers, scanning papers to images and then converting images to type II document through ORC software. This task, especially printing on papers, costs huge number of time and labor so is impossible when we need large amount of documents. So we give a substitute method: instead of printing on paper and converting them to image, we convert modified PDF document to images directly. Experiment has proved that type II document obtained from our substitute method has the same distortion as the real one.

We download 2104 type II pages (200 documents) from IEEE online library as our natural training samples. Using substitute method, we produce training stego groups with different μ and R (Tab. 4). Each group has 15 documents (150 pages in all). Classifier 3 is trained with extracted features which are described in Sec. 3.3. We download another 100 IEEE documents as our natural testing samples. To give a precise test on performance of classifier 3, we design two testing stego groups: variable shifting group and variable rate group (see Tab. 5).

Receiver Operating Characteristics (ROC) graph is a useful technique for visualizing the performance of classifiers (Ref. [14]). It is a graph in which true positive rate grows with false positive rate. To show our classifier's performance,

Table 4: training samples

μ (pt)	R	μ (pt)	R
0.5	10%	1	10%
0.5	30%	1	30%
0.5	50%	1	50%
0.5	70%	1	70%
1.5	10%	2	10%
1.5	30%	2	30%
1.5	50%	2	50%
1.5	70%	2	70%

Table 5: Two testing stego groups for classifier 3

(a) variable shifting group

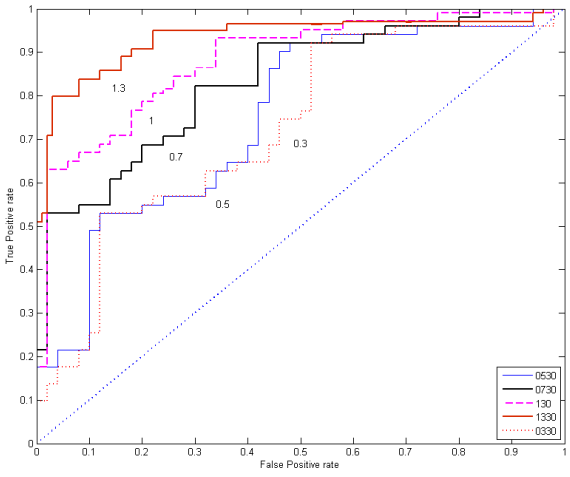
μ	R	ρ
0.3	30%	0.3494
0.5	30%	0.6110
0.7	30%	0.7862
1	30%	0.8107
1.3	30%	0.8312

(b) variable rate group

μ	R	ρ
1	5%	0.7962
1	20%	0.8084
1	35%	0.8112
1	55%	0.8204

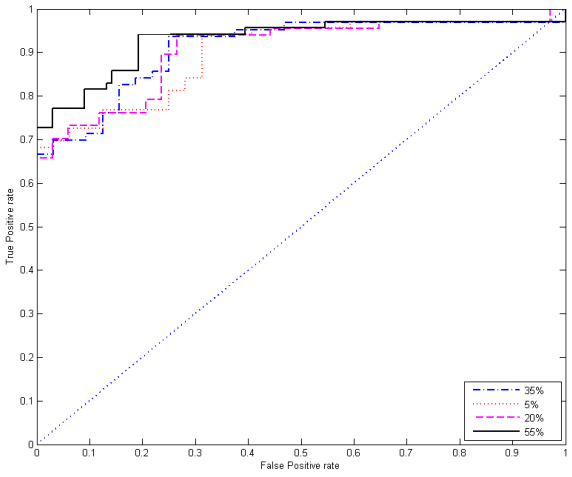
we firstly obtain octuple feature vectors (see Sec. 1) from the testing samples, then input them into probability SVM and calculate corresponding ROC curve from acceptable probability given by SVM. The result is shown in Fig. 6 and Fig. 7.

Figure 6: ROC graph of variable shifting group



In ROC analysis, area under ROC curve (AUC) is the measurement of classifier performance. The larger AUC ROC curve contain, the better the classifier performs. From the Fig. 7 and Fig. 6, we can see that our Classifier 3 has a

Figure 7: ROC graph of variable rate group



high performs in the tests because the AUC values are large. As defined in Ref. [15], the *detection reliability* ρ is:

$$\rho = 2A - 1$$

where A is the AUC. We list them in last column of Tab. 5. Fig 6 shows that when word shift modification grows larger, the classifier's detection efficiency goes better. In other words, steganographic scheme which shifts words far away from its origin position will have a high risk to be detected out. Notice the diagonal line in the graph, it represents the classifier using randomly guess ($\rho = 0$). When μ is 0.3 pt, our classifier performs worse, but it is much better than randomly guess. When μ is greater than 0.7 pt, our classifier will give a more trusty detection. From Fig 7, one can see that embedding rate R have much less influence on performance of classifier, compared with μ. When R varies from 5% to 55%, AUC does not increase much.

At last, we use another 114 natural documents (1018 pages) and 107 stego documents (973 pages) mixed with type I and type II to test our detection system. Two parameters of these stego documents μ and R are random values: $\mu \in [-2, +2]$, $R \in [1\%, 100\%]$. The detection result is shown in confusion matrix (Tab. 6)

Table 6: detection performance of our system

	Type I		Type II	
	nature	stego	nature	stego
right	42	50	61	49
wrong	3	4	8	8
total	45	54	69	57
accuracy	93.3%	92.5%	85.4%	85.9%

Experiments show that our classifier system can detect out stego PDF document with a satisfied accuracy. Documents with small word disturbance and low embedding rate ($\mu = 0.3, R = 5\%$) can also be detected out.

5. CONCLUSION

Text steganography has a potential application value because of wide use of text document on today's Internet. Therefore, its steganalysis is required to resist the illegal covert commmunication. In this paper, we have proposed a blind steganalysis method to detect the word shift steganographic algorithm in PDF documents. This method makes use of machine classifying methodology and employs SVM as our classifier. First, we divide PDF documents into two classes, "soft document" and "hard document", and establish our two-level detection model. Secondly, we design features specific to three detection classifiers. At last, a series of experiments prove that our method can detect out the word shift in PDF document with a satisfied accuracy.

This steganalytic method makes proposed word shift algorithm unsafe. Although we can not get the length or content of the secret message, detecting existence will lend much support for the next step analysis. If word shift steganography wants to resist our detection, it has to avoid modifying the statistical features which is used in our method. However, this will severely shrink the capacity of steganography. Hider needs to design more advanced text steganography if he still wants to use text in PDF document as covert communication channel.

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